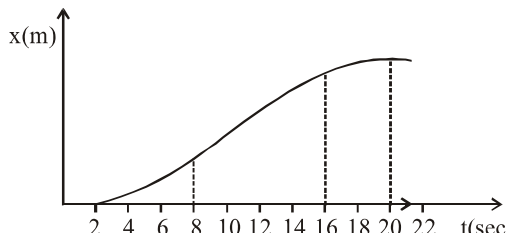


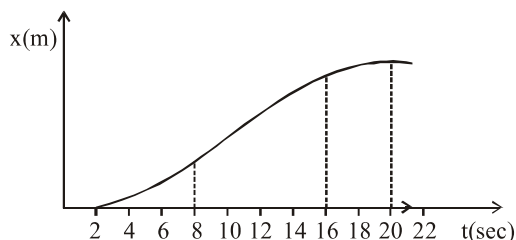
MOTION IN A STRAIGHT LINE

1. The location of a particle is changed. What can we say about the displacement & distance covered by the particle?
 - (1) Both cannot be zero
 - (2) One of the two may be zero
 - (3) Both must be zero
 - (4) Both must be equal
2. For a moving object, the numerical value of the ratio of magnitude of instantaneous velocity to instantaneous speed is:
 - (1) Always less than 1
 - (2) Always equal to 1
 - (3) Always more than 1
 - (4) Equal to or less than one
3. If the distance covered is zero, then displacement
 - (1) Must be zero
 - (2) May or may not be zero
 - (3) Can not be zero
 - (4) Depends upon the particle
4. A particle is moving in influence of constant acceleration. Its path may be :
 - (1) Circle
 - (2) Hyperbola
 - (3) Parabola
 - (4) Ellipse
5. Which of the following statement(s) is/are wrong:
 - (A) A body can have constant speed & varying acceleration
 - (B) A body can have zero velocity & uniform acceleration
 - (C) A body can have constant velocity & uniform acceleration
 - (1) Only C
 - (2) Both B & C
 - (3) Both A & C
 - (4) All A, B, C
6. Two bodies of masses 1 kg & 10 kg are dropped simultaneously from the top of a tower. On the surface of earth :
 - (1) Heavier will reach first
 - (2) Lighter will reach first
 - (3) Both will reach simultaneously
 - (4) Any one will reach first
7. A body is thrown upward & reaches its maximum height. At this position :
 - (1) Its velocity is zero & its acceleration is also zero
 - (2) Its velocity is zero but its acceleration is maximum
 - (3) Its acceleration is minimum
 - (4) Its velocity is zero & its acceleration is non zero due to gravity
8. Mark the correct statement :
 - (1) If velocity & acceleration have opposite sign, then speed of particle will increase
 - (2) If velocity is zero at an instant, then acceleration of particle will be zero at that instant
 - (3) If velocity is zero in a time interval then acceleration is not equal to zero at any instant within the time interval
 - (4) A particle moves along x-axis. If its x-coordinate & velocity have opposite sign then the particle will move towards origin
9. Which of the following statements is wrong about a ball thrown vertically up :
 - (A) It is moving with constant acceleration
 - (B) It may have different velocities at the same position
 - (C) It may have two positions at the same time
 - (D) The angular momentum of particle about point of projection remain conserved
 - (1) only C
 - (2) C, D
 - (3) B, C, D
 - (4) A, B, C, D
10. For uniform motion, the ratio of average velocity & instantaneous velocity is always :
 - (1) Less than 1
 - (2) Greater than 1
 - (3) Equal to 1
 - (4) None
11. 

What is nature of acceleration from $t = 2$ sec to $t = 10$ sec in the graph shown in above diagram

 - (1) Positive
 - (2) Negative
 - (3) Zero
 - (4) None of these

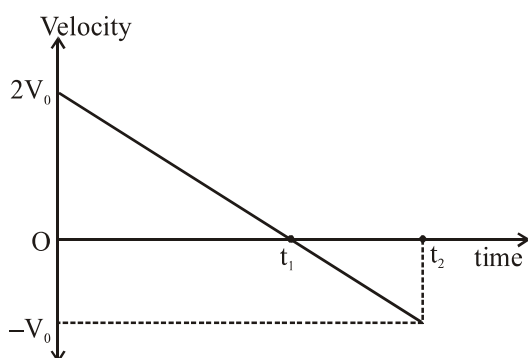
12.



What is nature of acceleration between $t = 18$ s to $t = 20$ s in above diagram

- (1) Positive (2) Negative
(3) Zero (4) None of these

13. In the given graph which of following physical quantities changes their sign?

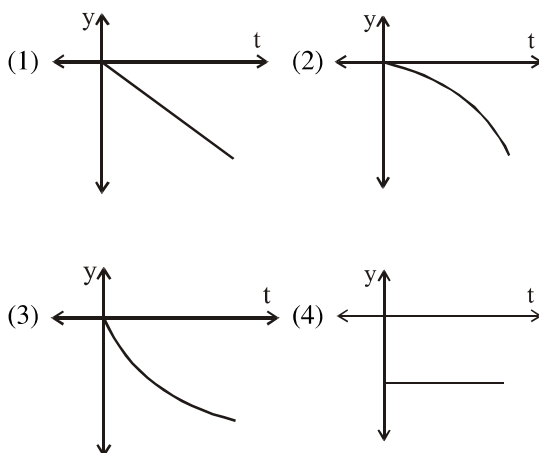


- (1) Displacement (2) Velocity
(3) Both (1) and (2) (4) Acceleration

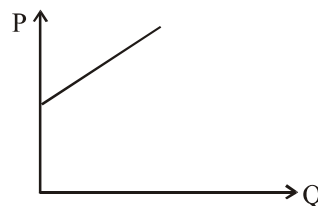
14. A ruler is dropped vertically through the gap between your thumb and fore finger. Then find reaction time if displacement of ruler is 21.0 cm?

- (1) 0.2 sec (2) 0.6 sec
(3) 0.8 sec (4) 0.9 sec

15. Which of following graph represents displacement-time of an object under free fall. Neglect air resistance.



16. Variation of quantity P with quantity Q is plotted in given figure, describes motion of particle in a straight line. Then which of the following statement is wrong:

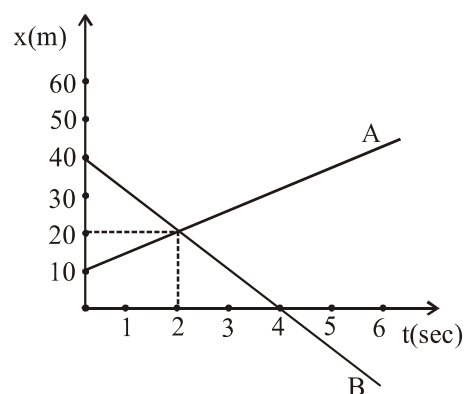


- (1) Quantity Q may represent time
(2) Quantity P is velocity if motion is uniform
(3) Quantity P may be displacement if motion is uniform
(4) Quantity P may be velocity if motion is uniformly accelerated

17. A ball is dropped from a building of height 45m. Simultaneously another ball is thrown up with a speed of 40 m/s. Calculate the magnitude of relative velocity of the balls when both balls are in air :

- (1) 0 m/s (2) 20 m/s
(3) 40 m/s (4) 60 m/s

18.



Displacement-time curve of two objects A & B is shown in the figure. Find relative velocity of B with respect to A

- (1) - 15 m/s (2) - 10 m/s
(3) 15 m/s (4) 10 m/s

19. A lift coming from 8th floor is just about to reach 4th floor and stops on 4th floor. Taking ground as origin and taking upward direction as positive for all quantities, which of following is correct?

- (1) $x < 0$, $v < 0$, $a < 0$
(2) $x > 0$, $v < 0$, $a < 0$
(3) $x > 0$, $v < 0$, $a > 0$
(4) $x > 0$, $v > 0$, $a < 0$

20. A car is moving on a road, then correct statement is
 (1) Car must be in motion for all observers
 (2) Car must be at rest for all observers
 (3) Car may be at rest or in motion, depends upon observer
 (4) Motion is independent upon the choice of observer
21. When a particle is released under gravity, then rate of change of velocity with respect to distance is :
 (1) Constant
 (2) Increases continuously
 (3) Decreases continuously
 (4) First increases then decreases
22. Number of possible directions of motion in one-dimensional motion is/are :
 (1) Only one (2) Only two
 (3) Only three (4) None of these
23. The relation $x = v_0 t + \frac{1}{2} a t^2$ is same as
 (Here $\bar{v}$ is average velocity)
 (1) $x = \left(\frac{v + v_0}{2} \right) t$ (2) $x = \frac{\bar{v} t}{2}$
 (3) $x = \left(\frac{v - v_0}{2} \right) t$ (4) $x = \left(\frac{\bar{v} + v_0}{2} \right) t$
24. For an object in free fall
 (1) Only gravitational force should act
 (2) Air resistance is neglected
 (3) Speed may or may not be zero
 (4) All of the above
25. Two objects are moving on straight line with non-zero velocities in opposite directions, then magnitude of relative velocity will be
 (1) Equal to individual velocity
 (2) Less than individual velocities
 (3) Greater than magnitude of individual velocities
 (4) Greater or equal to magnitude of individual
26. Two particles A and B are moving on a straight line with the same speed. Which of the following statement is/are correct for the relative motion of the two particles :
 (1) The relative velocity v_{AB} or v_{BA} is zero only if they are moving in the same direction
 (2) If the particles are moving in opposite direction, the magnitude of v_{BA} or v_{AB} is twice than the magnitude of velocity of A or that of B.
 (3) The relative velocity v_{AB} or v_{BA} is always zero
 (4) Both (1) and (2)
27. The stopping distance is an important factor for road safety and depends on
 (1) Initial velocity
 (2) Deceleration
 (3) Braking capacity
 (4) All of these
28. The relation $\langle \vec{v} \rangle = \frac{\vec{u} + \vec{v}}{2}$ is valid for
 (1) Constant acceleration only
 (2) Non-uniform acceleration only
 (3) Uniform motion only
 (4) Uniform and non-uniform acceleration both
29. Which of the following statement is/are **INCORRECT** ?
 (I) For motion involving change in direction along the straight line, displacement may decrease.
 (II) Direction of acceleration is same as direction of velocity.
 (III) Average speed is, in general greater than or equal to the magnitude of the average velocity.
 (IV) Equation of motion is valid only in one-dimensional uniform accelerated motion
 (1) only I (2) only IV
 (3) II, IV (4) I, III, IV